Intended use: problem solving

- Evaluation
  - Evaluation of a proposed system design for the purpose of assessing its quality characteristics such as operational effectiveness, integrated system effectiveness, deployment readiness, performance, interoperability, and security
- Comparison
  - Comparing competitive systems designed to carry out a specified function, or comparing several proposed operating policies or procedures
- Prediction
  - Forecasting the behavior of a system under some projected set of conditions
- Sensitivity Analysis
  - Determining which of many factors are the most significant in affecting overall system behavior
- Optimization
  - Determining exactly which combination of factor levels will produce the optimal overall behavior of the system
- Ranking and Selection
  - Ranking a number of alternatives (e.g., operating policies) and selecting the best one

LVC taxonomy

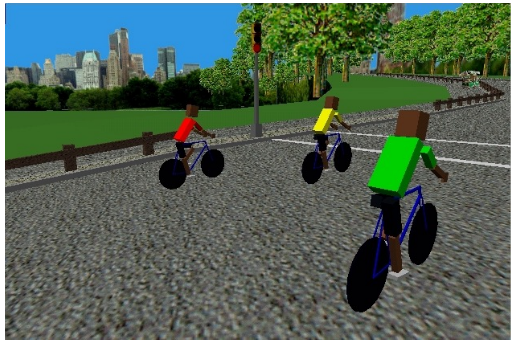
- Types of simulation
  - *Live*: real people operating real equipment (not computer-based)
  - *Virtual*: real people operating simulated equipment (often referred to as a "simulator," e.g., a flight simulator)
  - *Constructive*: simulated people operating simulated equipment

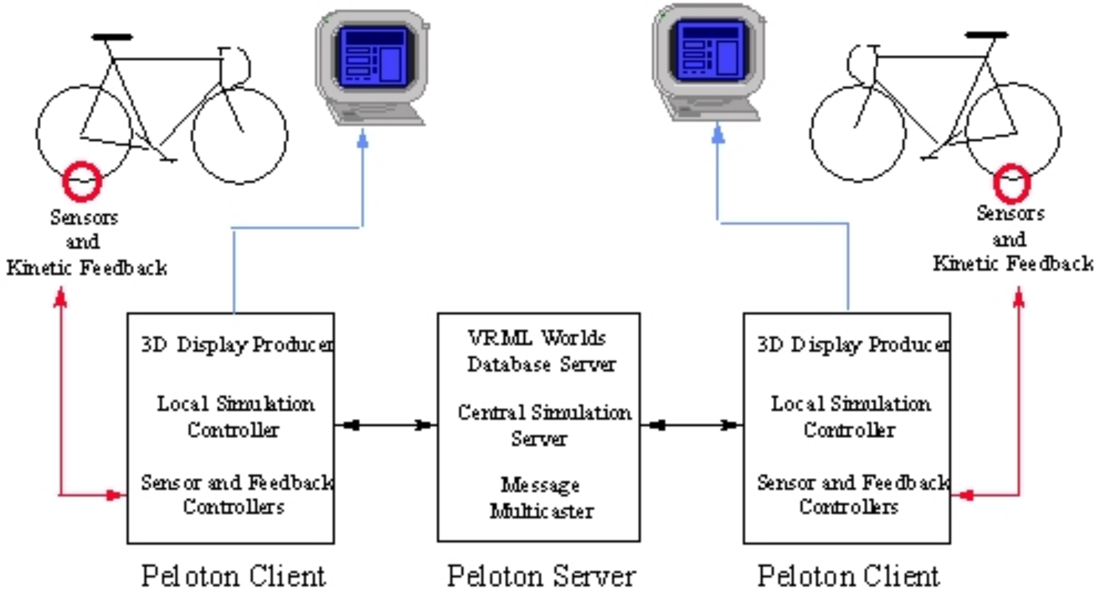
	Real Machines	**Simulated Machines**
Real People	Live	Virtual
Simulated People	Not defined *(hardware-in-the-loop)*	Constructive

- Major application areas
  - Analysis: logistics, operations
  - Training: platform level, command level
  - Test and evaluation: hardware-in-the-loop

Intended use: training

virtual simulation





Intended use: training

constructive simulation

Systems, models and simulation

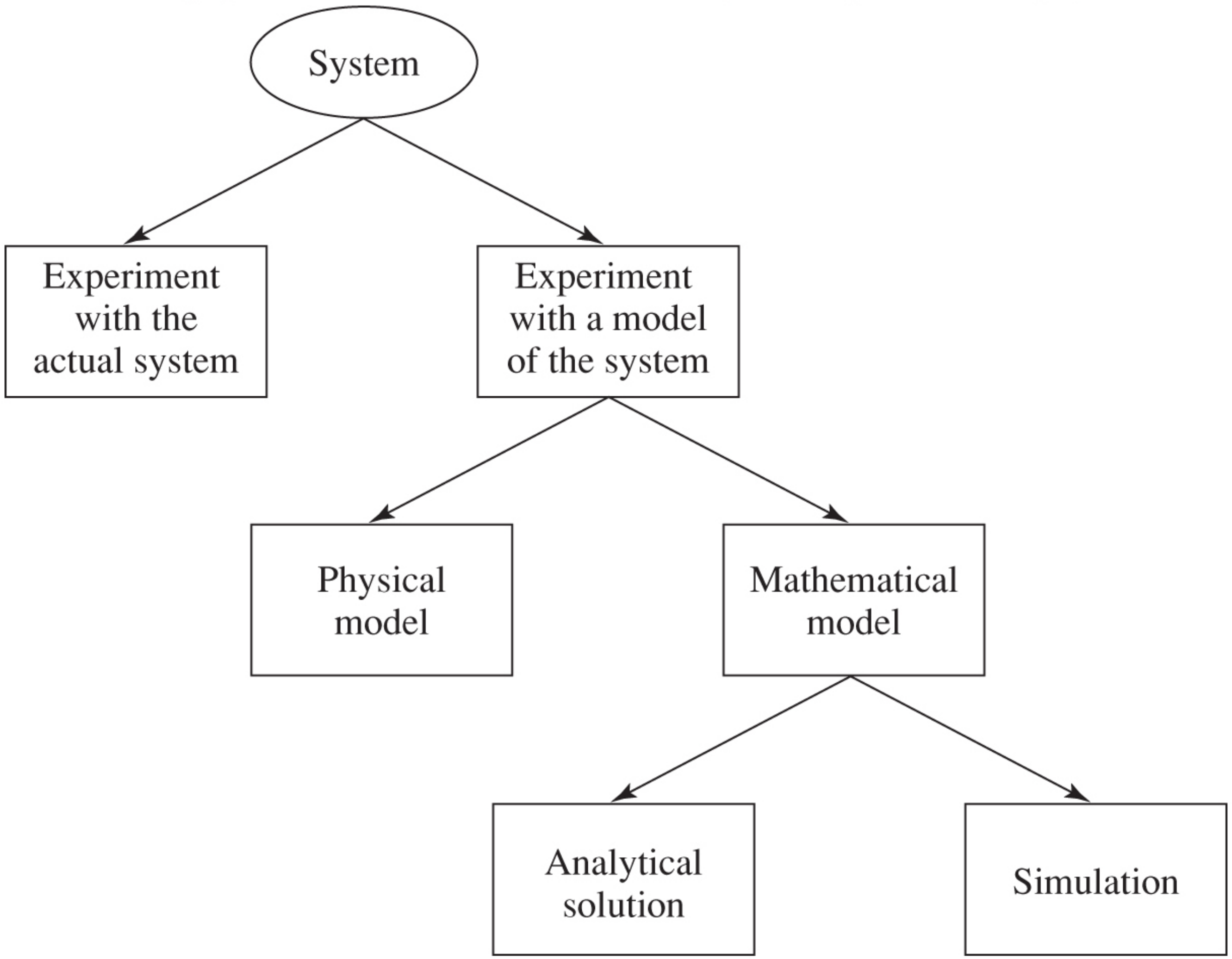
- System: A collection of entities (people, parts, messages, machines, servers, …) that act and interact together toward some end (Schmidt and Taylor, 1970)
  - In practice, depends on objectives of study
  - Might limit the boundaries (physical and logical) of the system
  - Judgment call: level of detail (e.g., what is an entity?)
  - Usually assume a time element – dynamic system
- State of a system: Collection of variables and their values necessary to describe the system at that time
  - Might depend on desired objectives, output performance measures
  - Bank model: Could include number of busy tellers, time of arrival of each customer, etc.

Systems, models and simulation

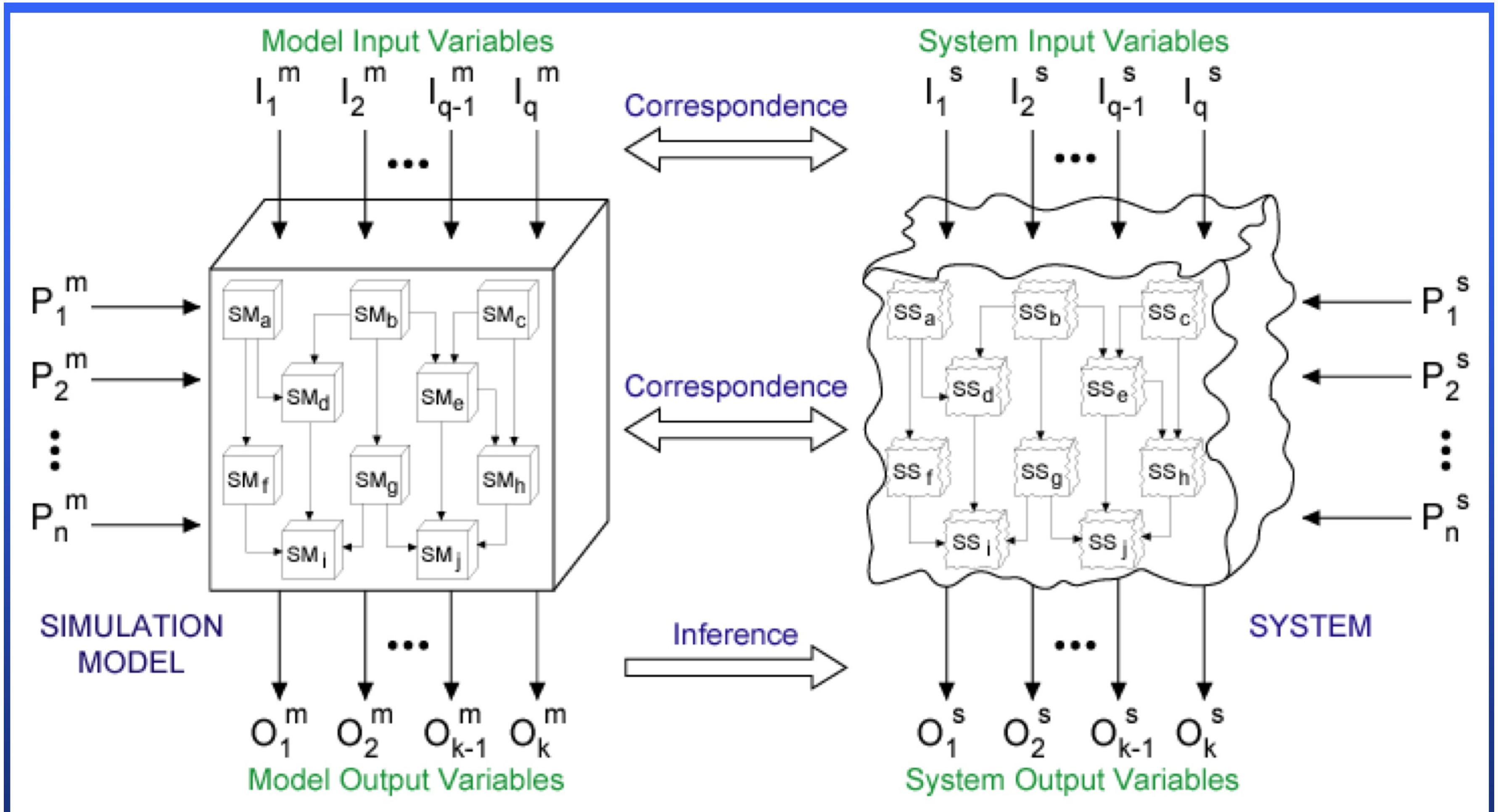
- Types of systems
  - Discrete
    - State variables change instantaneously at separated points in time
    - Bank model: State changes occur only when a customer arrives or departs
  - Continuous
    - State variables change continuously as a function of time
    - Airplane flight: State variables like position, velocity change continuously
- Most systems are *partly discrete, partly continuous*

Systems, models and simulation

- Ways to study a system



Systems and simulation models



I → Input
P → Parameter
O → Output

m → model
s → system
q, n, k → integer number

SM → Sub Model
SS → Sub System

Why simulation?

- Observing an operational system may be
  - Too expensive (e.g., computer chips)
  - Too dangerous (e.g., forest fires or chemical spills)
  - Too disruptive (e.g., traffic signal timing)
  - Too time consuming (e.g., weather)
  - Not possible (e.g., creation of the universe)
  - Morally or ethically unacceptable (e.g., spread of a disease)
  - Parts of the system may not be observable (e.g., internals of a silicon chip or biological system)

Why Simulation?

- One of the top technologies for *decision-making* and *analytics*
- Simulation is no longer the "technique of last resort"
- It is an indispensable problem-solving methodology
- Real systems are random
  - Demands
  - Time to failure
  - etc.
- Random systems cannot be easily analyzed with static modeling tools
- It is essential when analytical solutions are not feasible (i.e., the closed-form solution is not valid)

Types of problems

Problem Type	Given	Determine
Analysis (Direct)	Input, System	Output
Synthesis (Design Identification)	Input, Output	System
Instrumentation (Control)	System, Output	Input

Why Simulation?

- Modern systems are complex
  - Investment costs are high
  - Tolerance for error is low
- Simulation helps to
  - Hold costs down
  - Decrease errors (i.e., increase quality)
  - Save time and effort (i.e., avoid effort- and time-consuming rework activities)
- *Simulation* has today reached a level of predictive capability that it now firmly *complements the traditional pillars of science* (i.e., theory and experimentation/observation)

Historical overview

- 1940’s
  - Computer simulation in Manhattan Project to model nuclear detonation
- 1950’s
  - Analog computers commercially available and used for continuous simulations (solving differential equations)
- 1960’s
  - The digital computer offers an alternative to the analog computer and provides a means for simulating discrete event systems incorporating stochastic phenomena
  - SIMULA: the first object-oriented programming language
- 1970’s and 80’s
  - Numerous modeling and simulation applications are developed
- 1980’s and 90’s
  - emergence of parallel and distributed simulation (e.g., SIMNET, simulation standards such as DIS, ALSP, HLA, etc.)

Systems, models and simulation

- Classification of simulation models
  - Static vs. Dynamic
  - Deterministic vs. Stochastic
  - Continuous vs. Discrete
- Most operational models are dynamic, stochastic, and discrete
  - will be called discrete-event simulation models

Activities in a sound M&S study

the most critical, skills-requiring and error-prone activity